

Team members:

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Project: Marble game**Physics:**

RigidBodies: The player and moving obstacles use Rigidbody components to enable physics-based movement and interaction with the environment.

Force Application: Player movement is implemented using torque-based force. Instead of setting velocity, torque is applied to simulate a rolling marble.

Colliders: The player and other objects use colliders to detect and respond to physical interactions. For example, the player uses a sphere collider for smooth rolling. Obstacles and terrain use Box Colliders to detect when they are touched by the player.

Physics Materials: We use friction material for the player to allow smooth rolling. We also use the bounce material for some obstacles to make the player bounce from the obstacle.

Collision/Trigger Events: We use OnTriggerEnter() for collectibles and hazards, allowing detection. OnCollisionEnter() is used for obstacle impacts.

Reflection:

Wendy: I contributed to the creation of some maps, and implementing the spikes, bouncy obstacles, and coin collectibles. While working on these features, I learned to use tools such as Spline and ProBuilder. I used ProBuilder to create cones for the spikes since Unity does not

include that shape by default. Therefore, I had to install ProBuilder to have more variety of options for objects/shapes. I also learned Spline to create curved roads to make the game more visually appealing and dynamic. With that tool, I also learned that I could import examples to be able to use for the game. With that, I was able to implement the road. Overall, this process helped me learn how to use tools that can be useful for creating better games.

Taylor: I contributed to the implementation of the checkpoint system, the respawn system, the friction of the marble, the start and end screen, the timer, the next level logic, and the implementation of the textures and materials for some assets. As Unity is a new engine to me, all of this took a lot of trial and error and I learned a lot about C# and Unity as a whole. It was interesting to learn how to utilize features such as PlayerPrefs and integrating scripts within one another in order to make them all work together. I also learned that it's quite easy to connect the scenes using the SceneManager. I also had a lot of fun just messing around within Unity to do things like creating canvases. This project helped me realize that Unity is not as intimidating as it seems and it's actually quite easy to improve if you're willing to put in the time.